SCAFFOLD DESIGN REPORT

NIGERIA LIQUEFIED NATURAL GAS LIMITED

LOCATION: TRAIN 3	**AREA: D**
PERMIT TO WORK No.: —	**TYPE OF SCAFFOLD:** A Frame
SCAFFOLD DRAWING No: PMS-WA-ARCO-030	
RISK CLASSIFICATION: ○ High ✓ Medium ○ Low	

REFERENCE STANDARDS

- BS EN 1991-1-4:2005 — Wind Actions
- BS EN 1993-1-1:2005 — Design of Steel Structures (Eurocode 3)
- BS EN 12811-1:2003 — Temporary Works Equipment — Scaffolds

SCAFFOLD TO BE INSPECTED & TAGGED BEFORE USE

BRIEF DESCRIPTION OF SCAFFOLD DESIGN

This document describes the structural design of an **A Frame** scaffold required for *valve lifting* with equipment tag **311-UZ-202** . The scaffold system consists of a **1.8m × 1.8m × 3.5m** A-Frame scaffold with vertical bracing for sway resistance.

SAFE WORKING LOAD (SWL): 9.405 kN (959 kg)

TOTAL HORIZONTAL LOAD SUMMARY

Direction	Wind (kN)	Impact (kN)	TOTAL (kN)
X	2.49	2.822	**5.312**
Z	2.34	2.822	**5.162**

Role	ID	Name	Signature	Date
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TABLE OF CONTENTS

CALCULATION SHEETS

APPENDICES

GENERAL NOTE

BS EN 12811-1, 4.2.1.2

Scaffold materials are in accordance with BS EN 12811 with minimum design strength of 210 N/mm². This design calculation was carried out in accordance with the limit states design method stipulated in EN 1993-1-1:2005 (Eurocode 3 – Design of Steel Structures).

Material Properties

48.3 mm outside diameter × 3.6 mm thick steel tube was defined by STAAD.Pro Connect Edition V22 section tables (British) and assigned to all scaffold members as section **PIP483.6.**

Modelling Philosophy

BS EN 12811-1, 4.2.1.2

A 3-D model was analysed using **STAAD.Pro CONNECT Edition (V22)** in compliance with EN 1993-1-1:2005. Base supports are modelled as pinned with lateral spring stiffness representing frictional restraint (KFX/KFZ = 14.672 kN). Base supports are modelled as elastic spring supports, restrained laterally in both horizontal directions with rotational degrees of freedom released, representing flexible ground contact conditions at each standard foot. See Appendix A for model views.

LOAD CASES

Load diagrams are shown in Appendix B.

Dead Load / Self-weight

Self-weight of structural elements computed by STAAD.Pro with steel density **76.8195 kN/m³**, with a **10% allowance** for fittings and couplers (selfweight factor = 1.1).

Chain Hoist Load

BS EN 1991, 6.3.4.1

The equipment static weight is **767 kg ≈ 7.524 kN**. A **25% dynamic allowance** was applied for chain hoist operation.

> Static weight = 767 kg × 9.81 / 1000 = **7.524 kN**
> Chain Hoist Design Load (CL) = 7.524 × (1 + 0.25) = **9.405 kN**

Impact Load

BS EN 1991, 6.3.4.1

30% of the chain hoist design load was assumed as a horizontal impact load, applied at the hoist attachment point independently in X and Z directions.

> Impact Load (ILX) = 9.405 × 0.30 = **2.822 kN** (X-axis)
> Impact Load (ILZ) = 9.405 × 0.30 = **2.822 kN** (Z-axis)

Wind Load

EN 1991-1-4:2005

Wind loads were derived in accordance with EN 1991-1-4:2005 using NLNG Bonny Island site parameters. Full calculation chain is given in the Wind Load Calculation sheet below. Wind UDL = **0.055202 kN/m** applied as a member UDL to all exposed scaffold members in the global X and Z wind directions.

> Total Wind Load — X Axis (WLX) = **2.49 kN**
> Total Wind Load — Z Axis (WLZ) = **2.34 kN**

WIND LOAD CALCULATION — EN 1991-1-4:2005

Reference	Calculation	Value	Unit
	Basic wind speed (3-second gust)		
	V_3 =	**36.0**	m/s
	Eurocode $V_{b,0}$ is the characteristic 10-minute mean wind velocity		
Dredger P. — How to Approximate	Convert 3-second gust to 10-minute mean wind velocity		
	V_3 / V_{3600} =	1.52	
	V_{600} / V_{3600} =	1.06	
	$V_{600} = (V_{600}/V_{3600}) \times (V_{3600}/V_3) \times V_3$ =	**25.11**	m/s
	Basic wind speed $V_b = V_{600}$ =	**25.11**	m/s
EN 1991-1-4:2005	Roughness Length (z_0) — Terrain Category 0	0.003	m
	Terrain category II ($z_{0,ii}$)	0.05	
EN 1991-1-4, Table 4.1	z_{min}	1.0	m
	z — height of structure under consideration Source: STAAD command geometry height	**3.5**	m
EN 1991-1-4, Sec 4.3	$C_{r0}(z)$ — Roughness factor	1.0	
EN 1991-1-4, Sec 4.3	$C_o(z)$ — Orography factor	1.0	
EN 1991-1-4, Sec 4.3	Terrain factor $K_r = 0.19 \times [z_0/z_{0,ii}]^{0.07}$	0.156	
EN 1991-1-4, Table 4.1	$z_{min} < z \rightarrow$ Case 1 applies		
	Wind Turbulence		
	$C_r(z) = K_r \times \ln(z/z_0)$	1.1019	
	Mean velocity $V_m(z) = C_r(z) \times C_o(z) \times V_b$	27.664	m/s
EN 1991-1-4, Sec 4.4	K_1	1.0	
	Turbulence intensity $I_v(z) = k_1 / [C_o(z) \times \ln(z/z_0)]$	0.1416	
	Peak Velocity Pressure		
	Density of air ρ	1.25	kg/m³
EN 1991-1-4, Sec 4.5	$q_p(z) = [1 + 7 \cdot I_v(z)] \times 0.5 \times \rho \times V_m(z)^2$	**952.4103**	N/m²
Wind Load = $C_f \times q_p \times A_f$			
C_f =		1.2	
q_p =		0.95241	kN/m²
A_f =	Tube outer diameter (48.3 mm)	0.0483	m
Wind Load UDL =		**0.055202**	**kN/m**

LOAD COMBINATIONS

BS EN 12811-1, 6.2.9.2

Load combinations in accordance with BS EN 12811-1:2003. ULS combinations use γ = **1.5**; SLS (characteristic) combinations use γ = **1.0**.

DL = Dead Load | **CL** = Chain Hoist | **ILX/ILZ** = Impact X/Z | **WLX/WLZ** = Wind X/Z

LC No.	Combination	Component Loadings & Factors
7	**1.5DL + 1.5CL**	LC1 × 1.5 + LC2 × 1.5
8	**1.5DL + 1.5CL + 1.5ILX**	LC1 × 1.5 + LC2 × 1.5 + LC3 × 1.5
9	**1.5DL + 1.5CL + 1.5ILZ**	LC1 × 1.5 + LC2 × 1.5 + LC4 × 1.5
10	**1.5DL + 1.5WLX**	LC1 × 1.5 + LC5 × 1.5
11	**1.5DL + 1.5WLZ**	LC1 × 1.5 + LC6 × 1.5

LC No.	Combination	Component Loadings & Factors
12	**1.5DL + 1.5CL + 1.5WLX**	LC1 × 1.5 + LC2 × 1.5 + LC5 × 1.5
13	**1.5DL + 1.5CL + 1.5WLZ**	LC1 × 1.5 + LC2 × 1.5 + LC6 × 1.5
14	**DL + CL**	LC1 × 1.0 + LC2 × 1.0
15	**DL + CL + ILX**	LC1 × 1.0 + LC2 × 1.0 + LC3 × 1.0
16	**DL + CL + ILZ**	LC1 × 1.0 + LC2 × 1.0 + LC4 × 1.0
17	**DL + WLX**	LC1 × 1.0 + LC5 × 1.0
18	**DL + WLZ**	LC1 × 1.0 + LC6 × 1.0
19	**DL + CL + WLX**	LC1 × 1.0 + LC2 × 1.0 + LC5 × 1.0
20	**DL + CL + WLZ**	LC1 × 1.0 + LC2 × 1.0 + LC6 × 1.0

STATIC CHECK PARAMETERS — UTILIZATION RATIO SUMMARY

Maximum Utilization Ratio = 0.439 at Member No. **105**, Load Combination **7**, Clause **EC-6.2.9.1**. **PASS** — All members adequate; UC ratio below unity.

UC ratio diagram: see Appendix C.

Member	Section	Status	Clause	UC Ratio	L/C	F_x (kN)	Type
105	PIP483.6	PASS	EC-6.2.9.1	**0.439**	7	0.57	T
102	PIP483.6	PASS	EC-6.2.9.1	**0.415**	8	0.72	T
96	PIP483.6	PASS	EC-6.2.7(5)	**0.414**	8	8.70	C
94	PIP483.6	PASS	EC-6.3.3-662	**0.400**	9	3.00	C
108	PIP483.6	PASS	EC-6.2.9.1	**0.385**	8	0.03	T
93	PIP483.6	PASS	EC-6.3.3-662	**0.379**	9	2.86	C
104	PIP483.6	PASS	EC-6.3.3-662	**0.376**	7	8.85	C
31	PIP483.6	PASS	EC-6.3.3-662	**0.375**	8	0.12	C
103	PIP483.6	PASS	EC-6.2.9.1	**0.367**	7	8.71	T
90	PIP483.6	PASS	EC-6.3.3-662	**0.363**	9	2.74	C
75	PIP483.6	PASS	EC-6.2.9.1	**0.362**	8	8.64	T
79	PIP483.6	PASS	EC-6.2.9.1	**0.351**	8	1.30	C
46	PIP483.6	PASS	EC-6.2.9.1	**0.350**	8	1.62	T
112	PIP483.6	PASS	EC-6.2.9.1	**0.350**	8	1.62	T
99	PIP483.6	PASS	EC-6.2.9.1	**0.347**	9	0.50	T
73	PIP483.6	PASS	EC-6.2.9.1	**0.344**	8	3.01	T
30	PIP483.6	PASS	EC-6.2.9.1	**0.343**	8	0.95	T
72	PIP483.6	PASS	EC-6.2.9.1	**0.340**	9	3.04	T
32	PIP483.6	PASS	EC-6.2.9.1	**0.331**	8	0.07	T
80	PIP483.6	PASS	EC-6.2.9.1	**0.330**	9	1.16	C
89	PIP483.6	PASS	EC-6.3.3-662	**0.325**	9	2.64	C
69	PIP483.6	PASS	EC-6.2.9.1	**0.322**	9	3.06	T
74	PIP483.6	PASS	EC-6.2.9.1	**0.319**	8	1.27	T
82	PIP483.6	PASS	EC-6.2.9.1	**0.300**	8	1.12	C
83	PIP483.6	PASS	EC-6.2.9.1	**0.299**	9	1.21	C
28	PIP483.6	PASS	EC-6.2.9.1	**0.281**	8	0.09	C
68	PIP483.6	PASS	EC-6.2.9.1	**0.279**	9	3.44	T
59	PIP483.6	PASS	EC-6.3.3-662	**0.261**	8	1.38	C
110	PIP483.6	PASS	EC-6.2.9.1	**0.261**	7	4.52	C
38	PIP483.6	PASS	EC-6.3.3-662	**0.258**	8	0.21	C
95	PIP483.6	PASS	EC-6.2.9.1	**0.258**	8	1.38	C

Member	Section	Status	Clause	UC Ratio	L/C	F_x (kN)	Type
50	PIP483.6	PASS	EC-6.2.9.1	**0.249**	9	2.14	T
111	PIP483.6	PASS	EC-6.2.9.1	**0.249**	9	2.14	T
106	PIP483.6	PASS	EC-6.2.9.1	**0.246**	8	5.08	T
101	PIP483.6	PASS	EC-6.2.9.1	**0.243**	9	6.18	C
56	PIP483.6	PASS	EC-6.3.3-662	**0.222**	8	9.90	C
23	PIP483.6	PASS	EC-6.3.3-662	**0.221**	8	1.28	C
100	PIP483.6	PASS	EC-6.2.9.1	**0.218**	9	4.85	T
24	PIP483.6	PASS	EC-6.3.3-662	**0.215**	8	0.69	C
107	PIP483.6	PASS	EC-6.2.9.1	**0.213**	8	5.20	C
81	PIP483.6	PASS	EC-6.2.9.1	**0.202**	9	0.02	T
98	PIP483.6	PASS	EC-6.2.9.1	**0.201**	9	2.75	C
77	PIP483.6	PASS	EC-6.2.9.1	**0.198**	8	1.82	C
97	PIP483.6	PASS	EC-6.2.9.1	**0.195**	9	1.41	T
15	PIP483.6	PASS	EC-6.3.3-662	**0.189**	8	10.11	C
37	PIP483.6	PASS	EC-6.3.3-662	**0.189**	8	2.12	C
109	PIP483.6	PASS	EC-6.3.3-661	**0.188**	9	6.18	C
36	PIP483.6	PASS	EC-6.3.3-662	**0.176**	8	9.70	C
19	PIP483.6	PASS	EC-6.3.3-661	**0.171**	9	6.11	C
25	PIP483.6	PASS	EC-6.2.9.1	**0.169**	8	0.20	T
35	PIP483.6	PASS	EC-6.3.3-662	**0.166**	8	9.99	C
17	PIP483.6	PASS	EC-6.3.3-662	**0.165**	9	7.40	C
22	PIP483.6	PASS	EC-6.2.9.1	**0.164**	8	0.99	T
85	PIP483.6	PASS	EC-6.3.3-662	**0.164**	9	0.45	C
58	PIP483.6	PASS	EC-6.3.3-662	**0.162**	9	5.77	C
20	PIP483.6	PASS	EC-6.3.3-661	**0.154**	9	4.24	C
14	PIP483.6	PASS	EC-6.3.3-661	**0.150**	9	4.02	C
16	PIP483.6	PASS	EC-6.3.3-662	**0.149**	8	5.40	C
45	PIP483.6	PASS	EC-6.2.9.1	**0.146**	8	1.60	T
29	PIP483.6	PASS	EC-6.2.9.1	**0.144**	8	0.56	T
65	PIP483.6	PASS	EC-6.2.9.1	**0.138**	8	1.33	T
44	PIP483.6	PASS	EC-6.3.3-662	**0.137**	8	3.42	C
84	PIP483.6	PASS	EC-6.3.3-662	**0.128**	8	7.00	C
71	PIP483.6	PASS	EC-6.3.3-662	**0.125**	8	0.56	C
86	PIP483.6	PASS	EC-6.3.3-662	**0.121**	9	0.81	C
13	PIP483.6	PASS	EC-6.2.9.1	**0.120**	8	7.98	T
41	PIP483.6	PASS	EC-6.3.3-662	**0.120**	7	3.55	C
92	PIP483.6	PASS	EC-6.2.9.1	**0.119**	8	0.02	C
88	PIP483.6	PASS	EC-6.2.9.1	**0.117**	8	0.56	T
40	PIP483.6	PASS	EC-6.3.3-662	**0.114**	9	5.75	C
78	PIP483.6	PASS	EC-6.2.9.1	**0.112**	8	0.44	C
61	PIP483.6	PASS	EC-6.3.3-662	**0.108**	8	2.04	C
91	PIP483.6	PASS	EC-6.2.9.1	**0.107**	8	0.61	T
64	PIP483.6	PASS	EC-6.2.9.1	**0.106**	9	0.05	C
66	PIP483.6	PASS	EC-6.2.9.1	**0.106**	8	0.76	T
67	PIP483.6	PASS	EC-6.2.9.1	**0.102**	9	0.69	T
62	PIP483.6	PASS	EC-6.3.3-662	**0.098**	7	2.22	C
87	PIP483.6	PASS	EC-6.3.3-662	**0.096**	9	0.36	C

Member	Section	Status	Clause	UC Ratio	L/C	F_x (kN)	Type
121	PIP483.6	PASS	EC-6.3.3-662	**0.085**	8	1.76	C
43	PIP483.6	PASS	EC-6.2.9.1	**0.084**	7	0.36	T
70	PIP483.6	PASS	EC-6.2.9.1	**0.084**	8	0.40	T
49	PIP483.6	PASS	EC-6.2.9.1	**0.076**	9	2.14	T
113	PIP483.6	PASS	EC-6.3.3-661	**0.070**	8	3.21	C
115	PIP483.6	PASS	EC-6.3.3-661	**0.070**	9	3.97	C
116	PIP483.6	PASS	EC-6.3.3-661	**0.070**	9	3.98	C
55	PIP483.6	PASS	EC-6.3.3-662	**0.059**	7	0.18	C
114	PIP483.6	PASS	EC-6.3.3-662	**0.057**	9	2.52	C
118	PIP483.6	PASS	EC-6.3.3-661	**0.056**	8	3.07	C
117	PIP483.6	PASS	EC-6.3.3-661	**0.050**	8	2.72	C
76	PIP483.6	PASS	EC-6.2.9.1	**0.048**	8	0.17	T
57	PIP483.6	PASS	EC-6.3.3-662	**0.038**	12	0.61	C

SCAFFOLD TUBES CONNECTION STABILITY CHECK

EN 12811, Table C.1

The maximum axial load on scaffold tube members (from STAAD.Pro Connect Edition analysis) is **6.794 kN** (Member 96, C). **Right-angle double coupler of Class A (10 kN slipping resistance)** shall be used for all scaffold steel tube connections at node joints.

Check: 6.794 kN < 10.0 kN → PASS

Member force table: see Appendix D.

Table C.1 — Characteristic Values of Resistances for Couplers (EN 12811)

Coupler Type	Resistance	Characteristic Value	
		Class A	Class B
Right-angle Coupler (RA)	Slipping force $F_{s,k}$ (kN)	**10.0**	15.0
	Cruciform bending moment $M_{B,k}$ (kNm)	—	0.8
	Pull-apart force $F_{p,k}$ (kN)	20.0	30.0
	Rotational moment $M_{T,k}$ (kNm)	—	0.13
Swivel Coupler (SW)	Slipping force $F_{s,k}$ (kN)	6.0	9.0
	Cruciform bending moment $M_{B,k}$ (kNm)	—	0.8
	Pull-apart force $F_{p,k}$ (kN)	15.0	22.5
	Rotational moment $M_{T,k}$ (kNm)	—	0.08
Putlog / Single Coupler (PL)	Slipping force $F_{s,k}$ (kN)	—	3.0
	Cruciform bending moment $M_{B,k}$ (kNm)	—	0.6

FRICTIONAL RESISTANCE

Max vertical reaction per support (Max f_y) = Resistance × 0.15 / C_F	= **7.336 kN**
Coefficient of friction (C_F)	= **0.3**
Frictional Resistance (FR) = C_F × Max f_y	= **2.2008 kN**
Spring Stiffness = FR / 0.15	= **14.672 kN** ← applied as KFX/KFZ in STAAD model

DEFLECTION CHECK RESULTS

BS EN 12811-1, Cl 6.3 Serviceability limit state — values from STAAD.Pro SLS (unfactored) combination results. Diagrams and displacement tables are in Appendix C and Appendix D.

Vertical Deflection Check

Parameter	Value	Allowable	Check
Max vertical displacement — LC 15 Member 31 | Member length L = 900.0 mm	**0.581 mm**	L/100 = 900/100 = **9.0 mm**	PASS

Horizontal Deflection Check

Parameter	Value	Allowable	Check
Max horizontal displacement — LC 15 Member 38 | Member length L = 1450.0 mm	**1.349 mm**	L/200 = 1450/200 = **7.2 mm**	PASS

CONCLUSION

The structural analysis and design of the **A Frame** scaffold confirms the structure is adequate for its intended purpose. All member utilization ratios are below unity (maximum UC = **0.439** at Member 105). Vertical and horizontal deflections are within their respective allowable limits of **L/100 = 9.0mm** and **L/200 = 7.2mm**. All scaffold tube coupler connections are adequate for the applied axial loads.

RECOMMENDATIONS

The scaffold shall be inspected by a competent Scaffold Inspector after erection to confirm conformity with the design specification and approved working drawings before use. All erection work shall comply with the design intent and material specification. Any proposed change to the design requires a re-design and the Scaffold Engineer shall be duly informed.

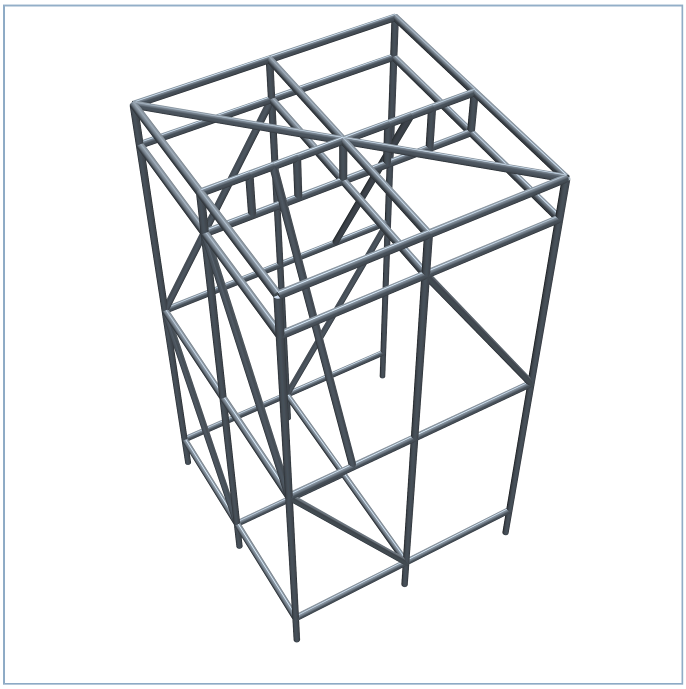
INSTALLATION NOTES

1. The scaffold working drawings and purpose of the scaffold must be reviewed and discussed by all personnel involved prior to execution.
2. The work vicinity must be barricaded and appropriate signage placed at all access points.
3. All scaffolders must wear complete PPE in accordance with NLNG work-at-height regulations.
4. A valid work permit must be obtained, discussed, and signed before erection commences.
5. Scaffold erection must be carried out in strict accordance with NASC guidance and site safety procedures.
6. Lay out and level the four base plates at the standard positions, ensuring a plan footprint of 1.8m x 1.8m in accordance with the approved working drawing.
7. Erect the four scaffold tube standards (48.3 x 3.6 mm) vertically on the base plates and secure with base collars. Check plumb in both axes before proceeding.
8. Install the lower horizontal ledgers and transoms at the kicker lift height (0.300m) to tie the four standards together and form the rigid base frame.
9. Install mid-level horizontal ledgers and transoms at 1.8m height in accordance with the working drawing.
10. Install mid-level horizontal ledgers and transoms at 3.2m height in accordance with the working drawing.
11. Fix all vertical bracing members diagonally across the face panels of the A-frame structure using right-angle couplers to provide sway resistance in both X and Z axes as shown on the working drawing.
12. Install the 1800mm ladder beam horizontally across the top of the standards at 3.5m height, connecting to both standards with right-angle double couplers. Ensure the beam is level before fully tightening all couplers.

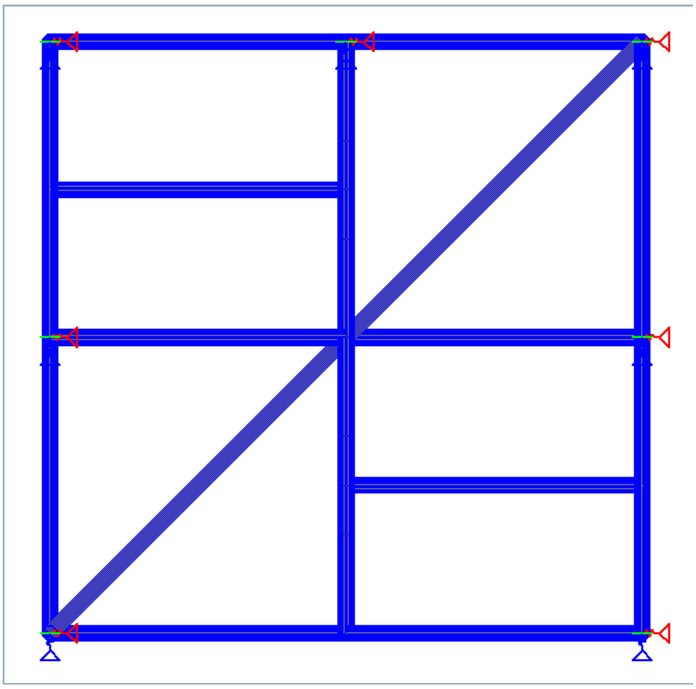
13. Rig the chain hoist to the ladder beam at the designated hoist point shown on the working drawing. Confirm the hoist is rated for a minimum safe working load equal to or greater than the design SWL.
14. The chain hoist attachment point and all coupler connections at ladder beam level shall be inspected and torque-checked by a competent person before any lifting operation commences.
15. The scaffold structure must be inspected and tagged by a competent advance scaffold inspector after erection and before any lift is performed.
16. The safe working load (SWL) stated on the design report must not be exceeded at any time during operations.
17. All modifications made after erection must be carried out by a competent scaffold team with the written approval of the scaffold engineer. End-user or third-party modification of this scaffold structure is strictly prohibited.
18. For dismantling, perform the erection steps in reverse order.

APPENDIX A — MODEL VIEWS

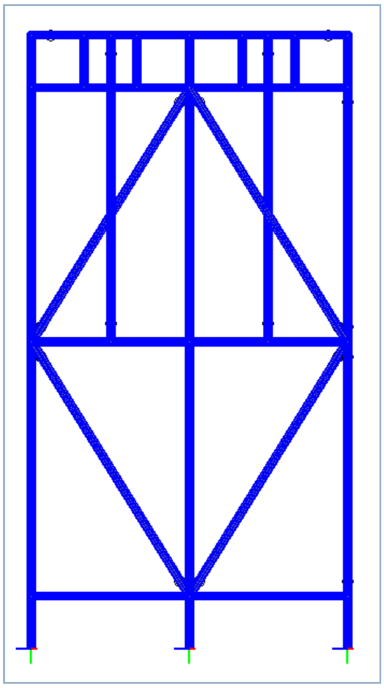
3D Isometric View



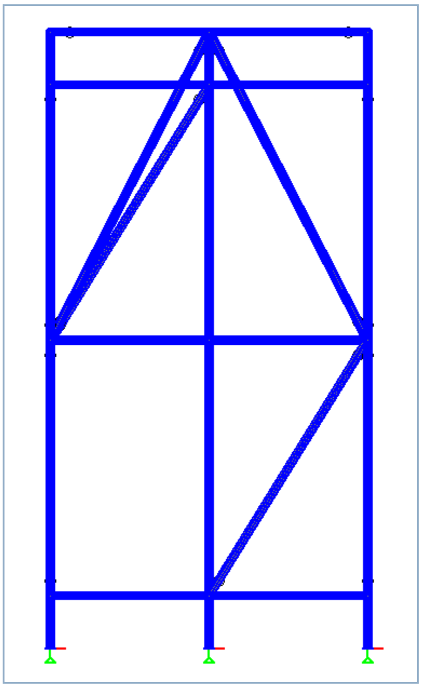
Plan View



Elevation from +X Axis



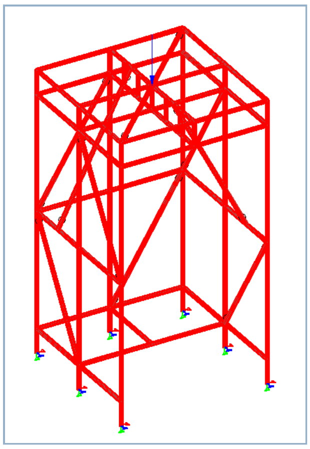
Elevation from +Z Axis



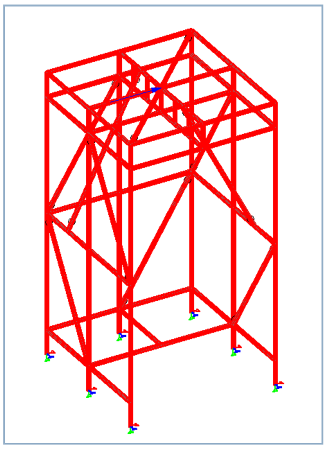
Scaffold Dimensions: W = 1.8m | D = 1.8m | H = 3.5m | **Lifts:** 0.3m, 1.75m, 3.2m, 3.5m

APPENDIX B — LOAD DIAGRAMS

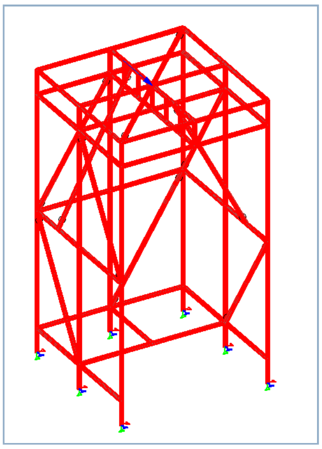
Chain Hoist Load (CL)



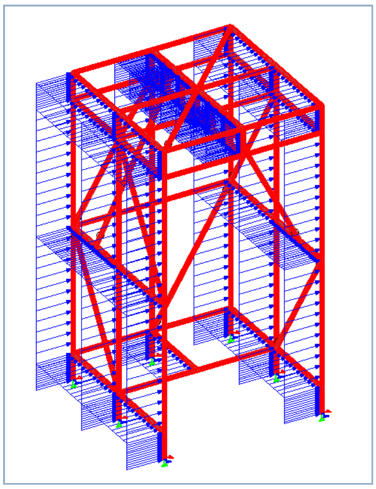
Impact Load — X Axis (ILX)



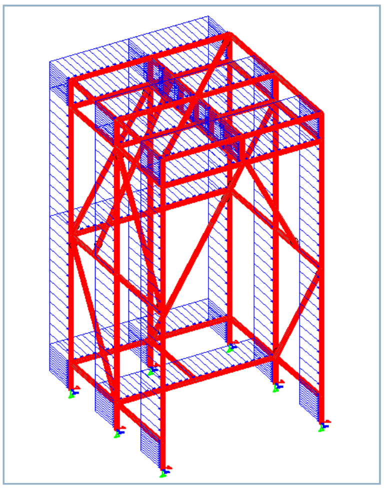
Impact Load — Z Axis (ILZ)



Wind Load — X Axis (WLX)

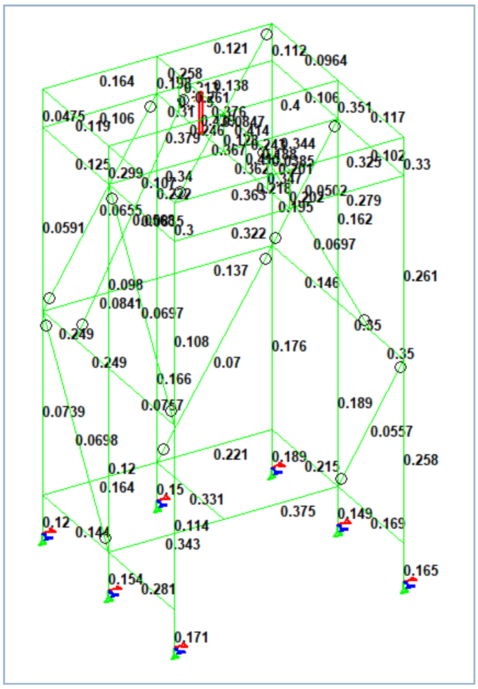


Wind Load — Z Axis (WLZ)

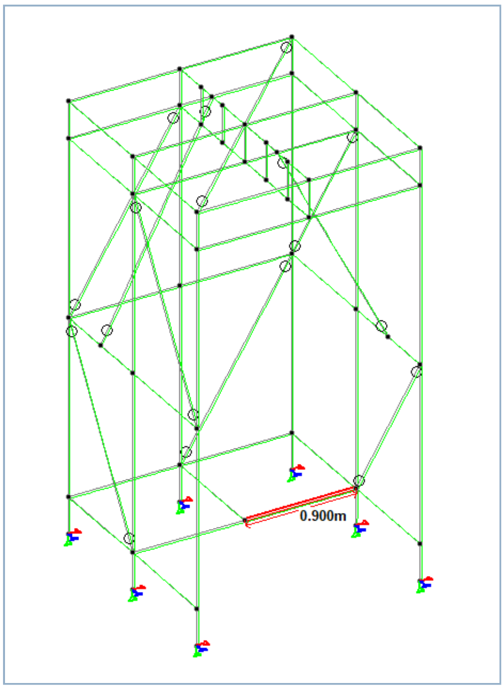


APPENDIX C — ANALYSIS RESULTS

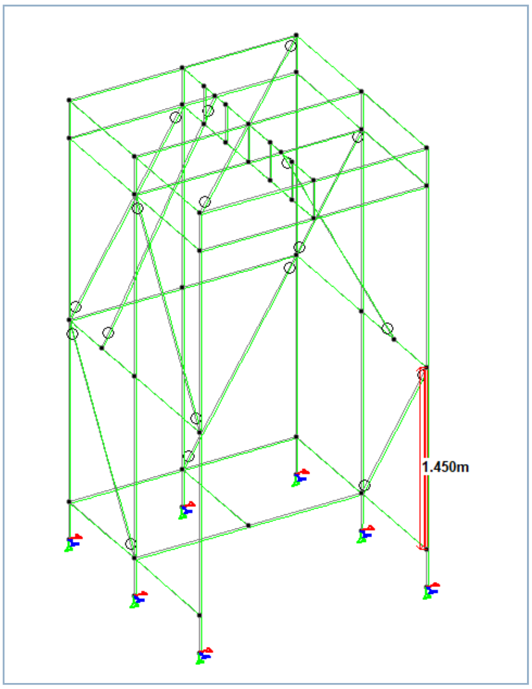
Utilization Ratio (UC) Diagram



Vertical Deflection Diagram



Horizontal Deflection Diagram



APPENDIX D — GUI RESULTS TABLES

Member Force / Code Check Table (Connection Stability)

Beam	L/C	Node	Axial Force kN	Shear-Y kN	Shear-Z kN	Torsion kN-m	Moment-Y kN-m	Moment-Z kN-m
96	16	45	6.794	-1.684	-0.233	-0.014	0.035	-0.340
104	15	51	6.107	1.667	0.219	-0.019	0.071	0.154
103	15	36	6.030	-1.631	-0.248	0.019	-0.139	0.336
104	20	51	5.903	1.848	-0.268	-0.014	0.042	0.181
75	16	48	5.901	1.700	0.068	0.015	0.011	-0.175
104	14	51	5.898	1.852	-0.275	-0.015	0.043	0.182
96	14	45	5.891	-1.571	-0.218	-0.013	0.032	-0.324
104	19	51	5.888	1.816	-0.235	-0.019	0.057	0.177
96	20	45	5.871	-1.569	-0.218	-0.013	0.032	-0.324
75	14	48	5.843	1.583	0.070	0.013	0.011	-0.157
96	19	45	5.841	-1.590	-0.248	-0.016	0.050	-0.326
75	20	48	5.834	1.582	0.068	0.014	0.011	-0.157
103	14	36	5.804	-1.863	0.063	0.014	0.005	0.365
96	15	45	5.798	-1.763	-0.716	-0.020	0.199	-0.353
103	19	36	5.798	-1.820	-0.029	0.017	-0.037	0.359
75	19	48	5.796	1.607	0.079	0.018	0.003	-0.162
75	15	48	5.757	1.817	0.217	0.020	-0.034	-0.197
103	20	36	5.747	-1.857	0.063	0.013	0.005	0.364
103	16	36	5.576	-1.711	0.069	0.013	0.006	0.340
104	16	51	4.788	1.699	-0.279	-0.014	0.044	0.161
109	16	54	4.123	-0.828	0.307	-0.003	-0.069	0.056
101	16	49	4.123	-1.981	-0.311	-0.003	-0.023	-0.242
110	15	55	3.464	1.826	0.134	-0.006	0.042	0.025
107	15	53	3.464	1.245	0.458	-0.006	-0.027	0.212
109	14	54	3.441	-0.753	0.297	-0.002	-0.066	0.052
101	14	49	3.441	-1.853	-0.294	-0.002	-0.022	-0.226

Vertical Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
31	0.581	0.525	15	1549	29.425	0.655	-2.745
30	0.460	0.300	15	1957	29.422	-0.219	3.317
31	0.456	0.525	19	1975	25.538	0.532	-1.324
28	0.456	0.525	15	1975	29.560	-0.006	4.405
30	0.388	0.375	17	2319	25.566	-0.375	1.519
31	0.373	0.600	17	2410	25.566	0.287	-1.696
30	0.320	0.375	19	2811	25.535	-0.170	1.470
28	0.283	0.525	19	3177	25.404	-0.020	2.129
25	0.271	0.375	15	3320	30.159	0.030	-4.428
28	0.266	0.525	17	3384	25.516	-0.002	2.269
94	0.257	0.225	16	3500	-0.203	-1.817	29.344
25	0.251	0.375	17	3592	25.561	-0.004	-2.256
73	0.248	0.225	16	3631	-0.204	-1.816	29.026
90	0.248	0.225	16	3634	-0.511	-1.662	29.178
73	0.235	0.225	15	3827	33.960	-1.719	-1.168
94	0.232	0.225	15	3887	34.065	-1.720	-0.990
69	0.229	0.225	16	3930	-0.483	-1.656	28.928
72	0.228	0.675	16	3942	-0.213	-1.781	28.851
93	0.228	0.675	16	3955	-0.194	-1.791	29.124
73	0.225	0.225	19	3992	27.950	-1.747	-0.584
68	0.225	0.675	16	4007	-0.473	-1.668	29.025
73	0.224	0.225	20	4018	-0.160	-1.764	23.457
94	0.224	0.225	20	4021	-0.167	-1.766	23.706
93	0.221	0.675	19	4067	27.932	-1.751	0.723
73	0.221	0.225	14	4077	-0.172	-1.749	0.061
94	0.218	0.225	14	4122	-0.184	-1.751	0.303
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Max = **0.581 mm** (LC 15) | L = Member 31 length (900 mm) | Allowable L/100 = 9.0 mm → **PASS**

Horizontal Displacement Table (SLS Combinations)

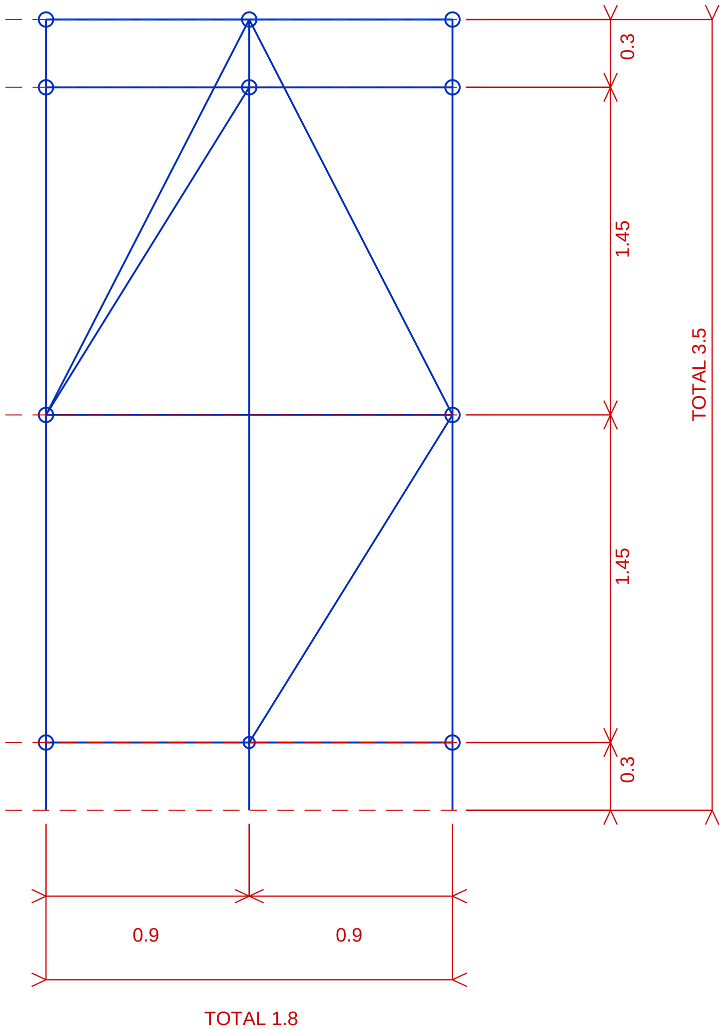
Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
38	1.349	0.846	15	1074	37.526	-0.002	-4.221
38	1.234	0.846	19	1174	31.288	-0.009	-2.000
38	0.719	0.725	17	2016	28.406	0.009	-2.272
59	0.714	0.483	15	2029	40.267	-0.007	-4.800
40	0.679	0.725	17	2136	28.128	-0.015	2.342
38	0.607	0.846	16	2387	2.579	-0.054	28.381
38	0.586	0.846	20	2474	2.496	-0.041	23.546
38	0.547	0.846	14	2648	2.536	-0.023	0.305
59	0.542	0.483	19	2672	32.613	-0.019	-2.373
40	0.526	0.725	19	2756	27.275	-0.027	2.170
37	0.496	0.967	19	2921	28.694	-0.024	-2.305
37	0.491	0.967	15	2954	33.429	-0.024	-4.677
58	0.485	0.483	15	2987	34.937	-0.032	-4.385
61	0.478	0.846	15	3035	37.361	-0.067	4.691
58	0.407	0.483	19	3558	29.412	-0.035	-2.094
40	0.402	0.725	15	3604	32.221	-0.033	4.526
59	0.369	0.604	17	3929	30.868	0.015	-2.409
61	0.356	0.604	17	4072	30.661	-0.028	2.424
56	0.310	0.967	14	4675	-0.019	-0.081	-0.222
56	0.305	0.967	19	4750	25.184	-0.122	-0.216
56	0.303	0.967	15	4783	28.007	-0.167	-0.191
59	0.296	0.363	16	4890	2.211	-0.088	28.083
56	0.292	0.967	20	4971	-0.015	-0.080	23.227
37	0.278	0.967	14	5215	1.281	-0.031	0.023
62	0.268	0.483	16	5417	-1.247	-0.011	27.774
37	0.263	0.967	16	5516	1.284	-0.051	28.014
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Max = **1.349 mm** (LC 15) | L = Member 38 length (1450 mm) | Allowable L/200 = 7.2 mm → **PASS**

APPENDIX E — REFERENCES & STANDARDS

Reference	Title	Year
BS EN 12811-1:2003	Temporary Works Equipment — Scaffolds — Performance Requirements and General Design	2003
BS EN 1993-1-1:2005	Eurocode 3: Design of Steel Structures — General Rules and Rules for Buildings	2005
BS EN 1991-1-4:2005	Eurocode 1: Actions on Structures — Wind Actions	2005
BS EN 1990:2002	Eurocode: Basis of Structural Design	2002
NASC TG20:13	Guide to Good Practice for Scaffolding with Tubes and Fittings	2013
Dredger P.	How to Approximate 10-Minute Mean Wind Speed from 3-Second Gust	—
STAAD.Pro CONNECT	STAAD.Pro CONNECT Edition V22 — Structural Analysis and Design Software	2022

Software Used: STAAD.Pro CONNECT Edition V22 (Bentley Systems) — 3D Structural Analysis and Code Checking per EN 1993-1-1:2005.
Report: A-Frame Scaffold Structural Design Report — ARCO M&E, NLNG Bonny Island.



FRONT VIEW

PROJECT
SCAFFOLDING WORKS

DRAWING NO.
PMS-WA-ARCO-003-FRONT-VIEW

DRAWING TITLE
SCAFFOLD GENERAL ARRANGEMENT - FRONT VIEW

SCALE	DATE	REV
NTS	**08-JUN-2026**	**0**

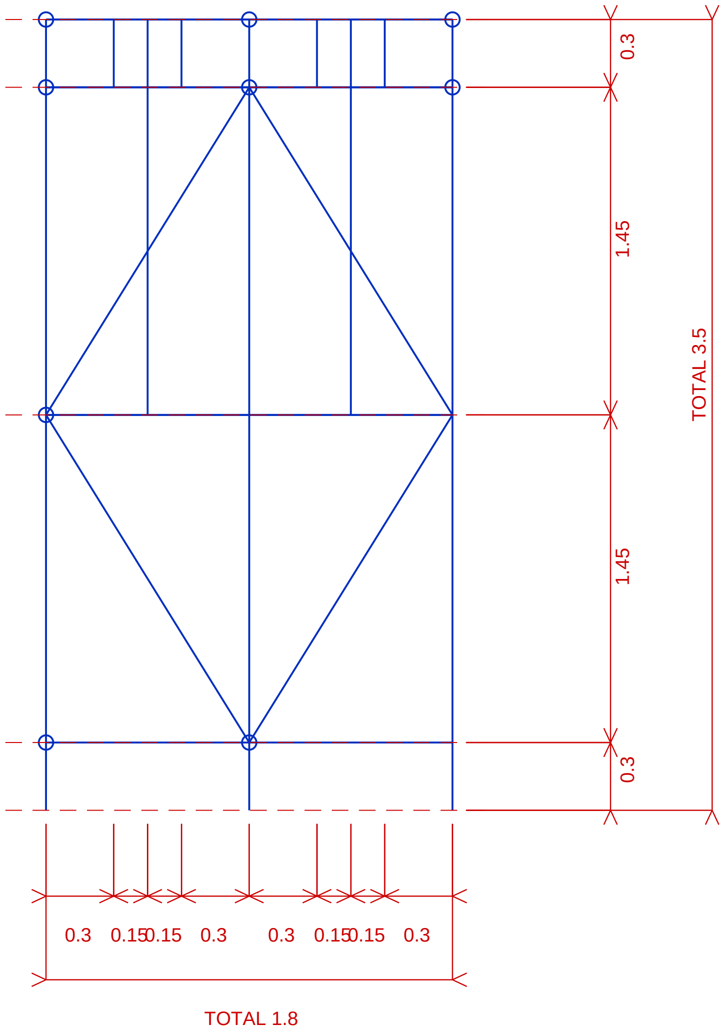
CLIENT	LOCATION	DRAWN	CHECKED	APPROVED
		Nnamani, Ifeanyi		

NLNG

ARCO M&E

COMPANY

SHEET	SIZE
1 OF 4	A4



SIDE VIEW

PROJECT
SCAFFOLDING WORKS

DRAWING NO.
PMS-WA-ARCO-003-SIDE-VIEW

DRAWING TITLE
SCAFFOLD GENERAL ARRANGEMENT - SIDE VIEW

SCALE	DATE	REV
NTS	**08-JUN-2026**	**0**

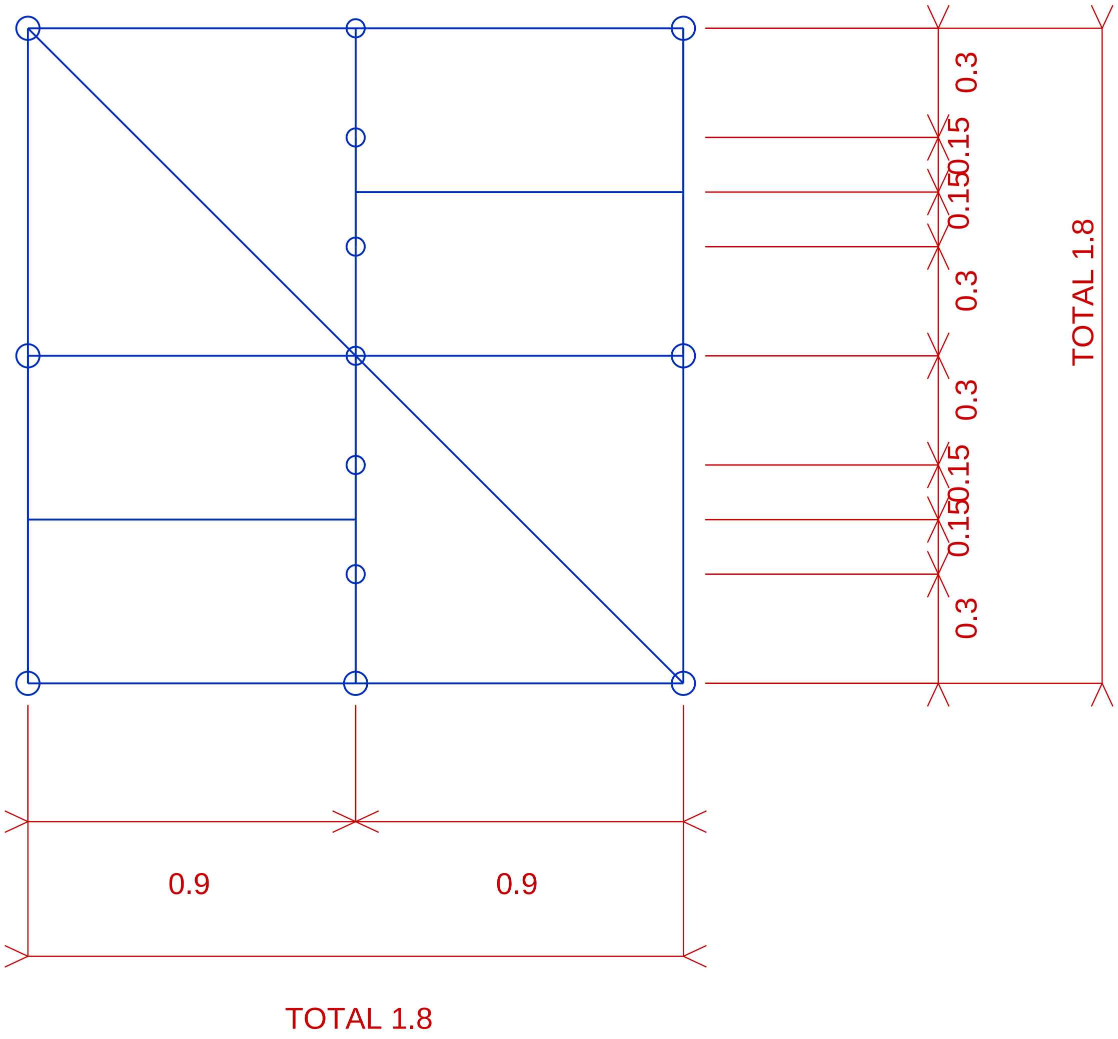
DRAWN	CHECKED	APPROVED
Nnamani, Ifeanyi		

CLIENT

LOCATION

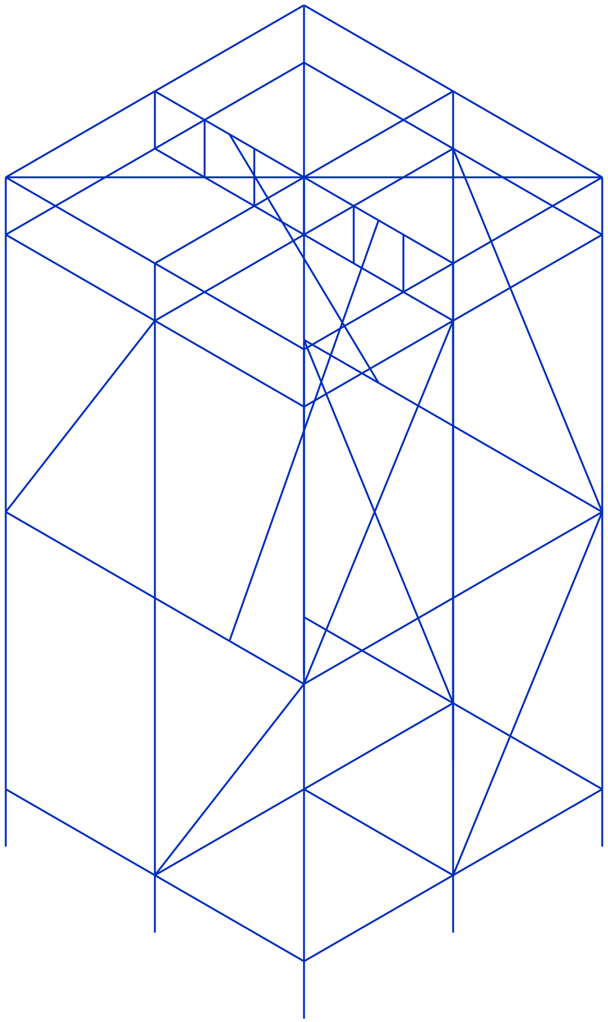
COMPANY

SHEET	SIZE
2 OF 4	A4



PLAN VIEW

PROJECT **SCAFFOLDING WORKS**		DRAWING NO. **PMS-WA-ARCO-003-PLAN-VIEW**			NLNG / ARCO M&E
DRAWING TITLE **SCAFFOLD GENERAL ARRANGEMENT - PLAN VIEW**		SCALE **NTS**	DATE **08-JUN-2026**	REV **0**	COMPANY
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED	SHEET **3 OF 4** / SIZE **A4**



3D VIEW

PROJECT **SCAFFOLDING WORKS**		DRAWING NO. **PMS-WA-ARCO-003-3D-VIEW**			NLNG / ARCO M&E
DRAWING TITLE **SCAFFOLD GENERAL ARRANGEMENT - 3D VIEW**		SCALE **NTS**	DATE **08-JUN-2026**	REV **0**	COMPANY
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED	SHEET **4 OF 4** / SIZE **A4**

3D Isometric View